WORKED SOLUTION

STEP III 1992, Question 8

Pure • Trigonometric Sums via Roots

STEP III 1992, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q8

The key idea

De Moivre's theorem turns $\sin(2n+1)\theta$ into a polynomial in $\cos\theta$ and $\sin\theta$, and dividing out $\sin^{2n+1}\theta$ leaves a polynomial in $\cot^2\theta$. The angles $k\pi/(2n+1)$ make the left-hand side vanish, so their squared cotangents are roots of a known polynomial. Once we know *all* the roots, the three sums are just Vieta's formulas (sums of roots and of reciprocals of roots) in disguise.

Solution

The identity

Write $c = \cos\theta$, $s = \sin\theta$ and $N = 2n+1$. By de Moivre's theorem and the binomial theorem,

$$\cos N\theta + \mathrm{i}\sin N\theta = (c+\mathrm{i}s)^N = \sum_{j=0}^{N}\binom{N}{j}c^{N-j}(\mathrm{i}s)^j.$$

The imaginary terms are those with j odd. Writing $j = 2m+1$, we have $\mathrm{i}^{2m+1} = \mathrm{i}(-1)^m$, so

$$\sin N\theta = \sum_{m=0}^{n}(-1)^m\binom{2n+1}{2m+1}c^{2n-2m}s^{2m+1}.$$

Now reverse the order of summation with $r = n-m$. Then $2m+1 = 2n+1-2r$, and $\binom{2n+1}{2n+1-2r} = \binom{2n+1}{2r}$ by the symmetry of binomial coefficients. So

$$\begin{aligned}\sin(2n+1)\theta &= \sum_{r=0}^{n}(-1)^{n-r}\binom{2n+1}{2r}c^{2r}s^{2n+1-2r}\\ &= \sin^{2n+1}\theta\sum_{r=0}^{n}(-1)^{n-r}\binom{2n+1}{2r}\cot^{2r}\theta,\end{aligned}$$

taking out s^{2n+1} and using $c^{2r}/s^{2r} = \cot^{2r}\theta$. This holds whenever $\sin\theta \neq 0$, as required.

The roots of the polynomial

Let

$$p(x) = \sum_{r=0}^{n} (-1)^r \binom{2n+1}{2r} x^r,$$

and put $\theta_k = k\pi/(2n+1)$ for $k = 1, \dots, n$. Then $(2n+1)\theta_k = k\pi$, so $\sin(2n+1)\theta_k = 0$, while $0 < \theta_k < \pi/2$ gives $\sin\theta_k \neq 0$. Dividing the identity by $\sin^{2n+1}\theta_k$ and multiplying by $(-1)^n$,

$$\sum_{r=0}^{n} (-1)^r \binom{2n+1}{2r} \cot^{2r}\theta_k = 0, \qquad \text{that is,} \qquad p(\cot^2\theta_k) = 0.$$

> **Ask yourself** How do we know these are *all* the roots? The sums below depend on it.

The angles $\theta_1 < \theta_2 < \cdots < \theta_n$ all lie in $(0, \pi/2)$, where cot is positive and strictly decreasing. So the n numbers $\cot^2\theta_k$ are distinct. The leading coefficient of p is $(-1)^n\binom{2n+1}{2n} = (-1)^n(2n+1) \neq 0$, so p has degree exactly n and so at most n roots. Hence the roots of $p(x) = 0$ are exactly $\cot^2\theta_k$ for $k = 1, \dots, n$, each simple.

The coefficients we need are

$$[x^n]:\ (-1)^n\binom{2n+1}{2n} = (-1)^n(2n+1),$$
$$[x^{n-1}]:\ (-1)^{n-1}\binom{2n+1}{2n-2} = (-1)^{n-1}\binom{2n+1}{3},$$
$$[x^1]:\ -\binom{2n+1}{2} = -n(2n+1), \qquad [x^0]:\ 1.$$

Part (i)

The sum of the roots is minus the ratio of the x^{n-1} and x^n coefficients:

$$\sum_{k=1}^{n} \cot^2\theta_k = -\frac{(-1)^{n-1}\binom{2n+1}{3}}{(-1)^n(2n+1)} = \frac{1}{2n+1}\cdot\frac{(2n+1)(2n)(2n-1)}{6} = \frac{n(2n-1)}{3}.$$

Part (ii)

Since $\tan^2\theta_k = 1/\cot^2\theta_k$, we want the sum of the reciprocals of the roots x_k. None of them is zero (the constant term is 1), and

$$\sum_{k=1}^{n} \frac{1}{x_k} = \frac{\sum_k \prod_{j\neq k} x_j}{\prod_k x_k},$$

whose numerator and denominator are, by Vieta, $(-1)^{n-1}\cdot[x^1]/[x^n]$ and $(-1)^n\cdot[x^0]/[x^n]$. Their ratio is $-[x^1]/[x^0]$, so

$$\sum_{k=1}^{n} \tan^2\theta_k = -\frac{-n(2n+1)}{1} = n(2n+1).$$

(Equivalently: the reciprocals $1/x_k$ are the roots of $x^n p(1/x)$, the polynomial with its coefficients reversed, and we read off its sum of roots.)

Part (iii)

Using $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$ and part (i),

$$\sum_{k=1}^{n} \operatorname{cosec}^2\theta_k = n + \frac{n(2n-1)}{3} = \frac{3n + 2n^2 - n}{3} = \frac{2n(n+1)}{3}.$$

Checks

For $n = 1$ the polynomial is $1 - 3x$, with root $\frac{1}{3} = \cot^2(\pi/3)$. Then (i) gives $\frac{1}{3}$, (ii) gives $3 = \tan^2(\pi/3)$ and (iii) gives $\frac{4}{3} = \operatorname{cosec}^2(\pi/3)$, all correct. For $n = 3$ the angles are $\pi/7, 2\pi/7, 3\pi/7$, and a calculator gives the three sums as 5, 21 and 8, matching $\frac{3\cdot5}{3}$, $3\cdot7$ and $\frac{2\cdot3\cdot4}{3}$. We checked the identity at random θ and n, and all three sums for $n \leq 29$, numerically.

Common slips

- Forgetting to argue that the $\cot^2\theta_k$ are distinct and that p has degree exactly n. Without this, Vieta's formulas can't be applied to the $\cot^2\theta_k$.
- Sign errors in the coefficients: the factor $(-1)^n$ cancels in part (i), but only if both coefficients carry it.
- Using k up to $2n$: the angles $k\pi/(2n+1)$ for $k > n$ repeat the same values of $\cot^2$, so only $k \leq n$ gives distinct roots.

Beyond STEP

Part (iii), with $\cot^2\theta < 1/\theta^2 < \operatorname{cosec}^2\theta$ for $0 < \theta < \pi/2$, squeezes $\sum_{k=1}^{n} 1/k^2$ between two multiples of the sums found here. Letting $n \to \infty$ gives Euler's famous result $\sum 1/k^2 = \pi^2/6$, a lovely elementary proof that first-year analysis courses often present. The same idea of reading sums off a polynomial's coefficients underlies symmetric functions and Newton's identities. For the analysis side, see Abbott, *Understanding Analysis*; for more problems in this spirit, Siklos, *Advanced Problems in Mathematics*.

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